Algebra SEP Worksheet 6 8/3/23

Field and Galois Theory

1. Give an example of a field F and an algebraic extension of F which has infinite degree.

1. Solution Consider the algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$. It is by definition an algebraic extension. If it had finite degree d, then let ζ_p be a primitive pth root of unity with $p-1>d$. Then $\mathbb{Q}(\zeta_p)$ has degree $p-1$ over $\mathbb{Q}$, so it can't be a subfield of $\overline{\mathbb{Q}}$, which is a contradiction. Thus, $\overline{\mathbb{Q}}/\mathbb{Q}$ has infinite degree.

2. (August 2017 Problem 4) Suppose that $K \subseteq \mathbb{C}$ is a Galois extension of $\mathbb{Q}$, $[K:\mathbb{Q}]=4$, and $\sqrt{-m} \in K$ for some positive integer m. Show that $\mathrm{Gal}(K/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

2. Solution This problem utilizes the same ideas as number 7 from the previous worksheet.

Since $K/\mathbb{Q}$ is Galois and degree 4, its Galois group is either $\mathbb{Z}/4\mathbb{Z}$ or $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. We can show that it is the latter by showing that K has two quadratic subfields, which means that $\mathrm{Gal}(K/\mathbb{Q})$ has two index 2 subgroups. We know that $\mathbb{Q}(\sqrt{-m}) \subseteq K$ is one quadratic subfield since $\sqrt{-m} \in K$ and the minimal polynomial of this element is x^2+m. Another quadratic subfield is given by the fixed field of complex conjugation. Let $c:\mathbb{C} \to \mathbb{C}$ be complex conjugation. Then c is a nontrivial automorphism of K since c is not the identity on the subfield $\mathbb{Q}(\sqrt{-m})$. Since c has order 2, the fixed field of K by c is a quadratic subfield of K different from $\mathbb{Q}(\sqrt{-m})$.

3. (January 2019 Problem 4)

(a) Let E be the splitting field over $\mathbb{Q}$ of $e(x)=x^3+2x^2+3x+4$, which has discriminant -200. Find $\mathrm{Gal}(E/\mathbb{Q})$.

(b) Let F be the splitting field over $\mathbb{Q}$ of $f(x)=x^3+4x^2+7x+6$, which has discriminant -72. Find $\mathrm{Gal}(F/\mathbb{Q})$.

(c) Find $\mathrm{Gal}(EF/\mathbb{Q})$.

3. Solution

(a) First, check to see if $e(x)$ has any rational roots. The possible rational roots are $\pm1, \pm2, \pm4$, none of work actually work. Since this is a degree 3 polynomial with no roots in $\mathbb{Q}$, this means that it is irreducible over $\mathbb{Q}$, so $\mathrm{Gal}(E/\mathbb{Q})$ is a transitive subgroup of S_3. So, it is either A_3 or S_3. Also, since $e'(x)=3x^2+4x+3$ has no real roots, $e(x)$ has no relative extrema, so $e(x)$ must just have one real root. This means that the other two roots are complex conjugates, so complex conjugation is an order 2 element of $\mathrm{Gal}(E/\mathbb{Q})$. Thus, $\mathrm{Gal}(E/\mathbb{Q})=S_3$.

Another way to get that $\mathrm{Gal}(E/\mathbb{Q}) \neq A_3$ is because $\mathrm{Gal}(E/\mathbb{Q}) \subseteq A_3$ iff the discriminant of $e(x)$ is a square in $\mathbb{Q}$. Since -200 is not a square, we get that $\mathrm{Gal}(E/\mathbb{Q}) \not\subseteq A_3$.

(b) Using the rational root theorem again, we find that -2 is a root of $f(x)$, and it factors as $f(x)=(x+2)(x^2+2x+3)$. Therefore, the roots of $f(x)$ are $-2, -1+\sqrt{-2}, -1-\sqrt{-2}$.

This gives that $F = \mathbb{Q}(\sqrt{-2})$, so $\mathrm{Gal}(F/\mathbb{Q}) = \mathbb{Z}/2\mathbb{Z}$.

(c) By definition, the discriminant of $e(x)$ is $[(\alpha_1-\alpha_2)(\alpha_1-\alpha_3)(\alpha_2-\alpha_3)]^2$ where $\alpha_1, \alpha_2, \alpha_3$ are the roots of $e(x)$. Therefore, the square root of the discriminant is an element in E. This gives that $\sqrt{-200} = 10\sqrt{-2} \in E$, so in particular, $F = \mathbb{Q}(\sqrt{-2}) \subseteq E$. Therefore, $EF = E$ and $\mathrm{Gal}(EF/\mathbb{Q}) = S_3$.

4. (January 2018 Problem 5) Let K be the splitting field of the polynomial $f(x) = x^4 - x^2 - 1$ over $\mathbb{Q}$. Compute $\mathrm{Gal}(K/\mathbb{Q})$. (For partial credit, find the degree $[K : \mathbb{Q}]$.)

4. Solution Since $f(x)$ has degree 4, we know that $|\mathrm{Gal}(K/\mathbb{Q})| \leq 4!$ and that $\mathrm{Gal}(K/\mathbb{Q})$ is a transitive subgroup of S_4. So, it could be $\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/2\mathbb{Z}$, D_4, A_4, or S_4. To find the roots of $f(x)$, notice that it is a quadratic in disguise. Letting $t = x^2$, we have that $f(t) = t^2 - t - 1$, which has roots $\alpha = \frac{1+\sqrt{5}}{2}$ and $\beta = \frac{1-\sqrt{5}}{2}$. Therefore, the roots of $f(x)$ are given by $x = \pm\sqrt{\alpha}, \pm\sqrt{\beta}$, and thus, $K = \mathbb{Q}(\sqrt{\alpha}, \sqrt{\beta})$.

Observe that K contains the quadratic subfield $\mathbb{Q}(\alpha) = \mathbb{Q}(\beta) = \mathbb{Q}(\sqrt{5})$, so the Galois group must contain an index 2 subgroup. Therefore, it cannot be A_4. Also, since $K = \mathbb{Q}(\sqrt{\alpha}, \sqrt{\beta})$ has degree at most 4 over $\mathbb{Q}(\sqrt{5})$, the Galois group of $K/\mathbb{Q}$ can have order at most 8. So, it cannot be S_4.

We will now show that $\mathrm{Gal}(K/\mathbb{Q}) = D_4$, a group of order 8, by showing that $[K : \mathbb{Q}] > 4$, which will eliminate $\mathbb{Z}/4\mathbb{Z}$ and the Klein 4 group as options. Consider the subfield $L = \mathbb{Q}(\sqrt{\alpha}, \beta) \subseteq K$, which is not all of K since $L \subseteq \mathbb{R}$ and $K \not\subseteq \mathbb{R}$. We will show that $[L : \mathbb{Q}] = 4$, which implies that $[K : \mathbb{Q}] > 4$. Since $\mathbb{Q}(\sqrt{5}) \subseteq L$, it is enough to show that L is strictly larger than $\mathbb{Q}(\sqrt{2})$. Suppose instead that $\sqrt{\alpha} = a + b\sqrt{5}$ with $a, b \in \mathbb{Q}$. Then squaring both sides gives

$$\frac{1}{2} + \frac{1}{2}\sqrt{5} = a^2 + 5b^2 + 2ab\sqrt{5}$$

which implies that $\frac{1}{2} = a^2 + 5b^2$ and $\frac{1}{2} = 2ab$. Substituting $b = \frac{1}{4a}$ into the first equation and simplifying (since $a \neq 0$) gives $16a^4 - 8a^2 + 5 = 0$. You can then use the rational root theorem to show that none of the possible roots in $\mathbb{Q}$ are actually roots. Therefore, $\sqrt{\alpha} \notin \mathbb{Q}(\sqrt{5})$, so $[L : \mathbb{Q}] = 4$, and we are done.

5. (August 2022 Problem 4) Let p be a prime number, $\mathbb{F}_p$ the finite field with p elements, and $K = \mathbb{F}_p(t)$ the field of rational functions in one variable t over $\mathbb{F}_p$. (So that K is the fraction field of the polynomial ring $\mathbb{F}_p[t]$).

(a) Show that the splitting field of the polynomial $x^p - t \in K[x]$ over K is inseparable.

(b) Show that the splitting field of $x^p - t - 1 \in K[x]$ over K is the same as the splitting field of $x^p - t$.

(c) Show that K has a single degree p inseparable extension.

5. Solution

(a) A field extension L/K is said to be *inseparable* if there exists some $\alpha \in L$ whose minimal polynomial is inseparable. In this case, let L be the splitting field of $x^p - t$, and let $\alpha = \sqrt[p]{t}$ be a root of $x^p - t$. The polynomial $x^p - t$ is irreducible over K by Eisenstein's criterion, so this is the minimal polynomial of $\sqrt[p]{t}$. However, as in Problem 8(b) on the previous worksheet, $x^p - t$ factors as $(x - \sqrt[p]{t})^p$, so this polynomial is not separable.

(b) Note that $(\sqrt[p]{t} - 1)^p = (\sqrt[p]{t})^p - 1^p = t - 1$, so $\sqrt[p]{t-1} \in K(\sqrt[p]{t})$ and visa versa, so these are the same extension.

(c) Let L/K be a degree p inseparable field extension. First, we want to show that $L = K(\alpha)$ for some inseparable element α. Certainly, L has an element α whose minimal polynomial is inseparable. Furthermore, since field degrees are multiplicative, we have $p = [L : K] = [L : K(\alpha)][K(\alpha) : K]$, and moreover $[K(\alpha) : K] \neq 1$ (or else $\alpha \in K$ is separable), so $[K(\alpha) : K] = p$ and thus $K(\alpha) = L$.

Now, it is a fact that any inseparable irreducible polynomial is of the form $g(x^p)$ for some irreducible polynomial $g(x)$. Thus, the only inseparable irreducible polynomials of degree p are of the form $x^p - f$ for some $f \in K$. But notice that $f(\sqrt[p]{t})^p = f(\sqrt[p]{t^p}) = f(t)$, so that $\sqrt[p]{f} \in K(\sqrt[p]{t})$. Thus, the only degree p inseparable extension of K is $K(\sqrt[p]{t})$.

6. (January 2022 Problem 5) Let F be a field of prime characteristic p. Suppose $E = F(\alpha)$ is a simple extension such that $\alpha \notin F$ but $\alpha^p - \alpha \in F$.

(a) Find $[E : F]$.

(b) Prove that E/F is a Galois extension.

(c) Find the Galois group $\mathrm{Gal}(E/F)$.

Hint: What is $(x + 1)^p - (x + 1)$ in characteristic p?

6. Solution

(a) Let $\alpha^p - \alpha = b \in F$, so that α is a root of the polynomial $f(x) = x^p - x - b$. Thus, $[E : F] \leq p$. We will show that $[E : F] = p$, but we will have to come back to it.

(b) Notice that $(x + 1)^p - (x + 1) = x^p - x$, so $\alpha, \alpha + 1, \ldots, \alpha + p - 1$ are the p roots of f. Thus, f splits in E. Furthermore, f clearly has no repeated roots, so f is separable. Thus, E is the splitting field of a separable polynomial, and thus is Galois.

(c) The Galois group $\mathrm{Gal}(E/F)$ has size at most p, since we have already seen that $[E : F] \leq p$. But it is easy to check that $\alpha \mapsto \alpha + i$ gives p different automorphisms of $\mathrm{Gal}(E/F)$, so this must be all of them. Thus, $\mathrm{Gal}(E/F) \cong \mathbb{Z}/p\mathbb{Z}$, generated by $\alpha \mapsto \alpha + 1$.

(a) (again) Since E/F is a Galois extension and $|\mathrm{Gal}(E/F)| = p$, we must have $[E : F] = p$.

7. (August 2015 Problem 5) Let $\mathbb{C}(x)$ be the field of rational functions with complex coefficients in the variable x. Thus, x is transcendental over $\mathbb{C}$. Put

$$y = x^n + x^{-n} \in \mathbb{C}(x)$$

for some $n > 0$. Prove that the field extension $\mathbb{C}(x)/\mathbb{C}(y)$ is a finite Galois extension and find its degree.

7. Solution Let ζ be a primitive nth root of unity, let $\sigma_i : \mathbb{C}(x) \to \mathbb{C}(x)$ be the automorphism induced by $x \mapsto \zeta^i x$ for $0 \le i \le n-1$, and let τ be the automorphism defined by $x \mapsto \frac{1}{x}$. Notice that all of the σ_i and τ fix $\mathbb{C}(y)$ since $\zeta^n = 1$, so their products $\sigma_i\tau$ also fix $\mathbb{C}(y)$. Therefore, $Id, \sigma_1, \ldots, \sigma_{n-1}, \tau, \sigma_1\tau, \ldots, \sigma_{n-1}\tau$ are all elements of $\operatorname{Aut}(\mathbb{C}(x)/\mathbb{C}(y))$, so $|\operatorname{Aut}(\mathbb{C}(x)/\mathbb{C}(y))| \ge 2n$. On the other hand, we have that x is a root of $t^{2n}+1-(x^n+x^{-n})t^n \in \mathbb{C}(y)[t]$, which is a degree $2n$ polynomial, so $[\mathbb{C}(x) : \mathbb{C}(y)] \le 2n$. Since $|\operatorname{Aut}(\mathbb{C}(x)/\mathbb{C}(y))| \le [\mathbb{C}(x) : \mathbb{C}(y)]$, they must both be equal to $2n$. Thus, $\mathbb{C}(x)/\mathbb{C}(y)$ is a Galois extension of degree $2n$.

8. (August 2020 Problem 3) Let L/K be a degree p Galois extension, where p is prime, and let σ be a nontrivial element of $\operatorname{Gal}(L/K)$. Then σ is a linear transformation of the K-vector space L.

(a) What is the characteristic polynomial of σ?

(b) Write L^0 for the subspace of L consisting of elements with trace 0. Show that σ sends L^0 to L^0.

(c) What is the characteristic polynomial of σ acting on L^0?

8. Solution

(a) Since $|\operatorname{Gal}(L/K)| = [L : K] = p$, $\operatorname{Gal}(L/K)$ must be the cyclic group of order p. So, in particular, σ satisfies $x^p - 1 = 0$. Furthermore, σ in fact generates $\operatorname{Gal}(L/K)$, so $\mathrm{id}, \sigma, \sigma^2, \ldots, \sigma^{p-1}$ are all the automorphisms of L/K. By linear independence of characters, these automorphisms are all linearly independent over K, so σ does not satisfy any polynomial of degree $< p$ over K. Hence, $x^p - 1$ is both the minimal and characteristic polynomial of σ.

(b) Recall that the trace of an element is the sum of its Galois conjugates. So, if $\operatorname{Tr}(\alpha) = 0$, then we have $\operatorname{Tr}(\sigma(\alpha)) = \sum_{g \in \mathrm{Gal}} g(\sigma(\alpha)) = \sum_{g' \in \mathrm{Gal}} g'(\alpha) = 0$. Here we have used that the action of a group on itself by right multiplication is free.

(c) Since $\operatorname{Tr}(\alpha) = 0$ is a single K-linear condition, we know that L^0 is a dimension $p-1$ vector space over K. So, we're looking for a degree $p-1$ polynomial, and moreover we know that polynomial has to divide $x^p - 1$. Now, we have to split into cases based on the characteristic. If $\operatorname{char}(K) = p$, then we know $x^p - 1 = (x-1)^p$, and so if we're trying to find a degree $p-1$ polynomial dividing it we have only one choice: $(x-1)^{p-1}$.

If $\operatorname{char}(K) \ne p$, we've gotta figure out which root of $x^p - 1$ is not an eigenvalue of σ acting on L^0. And let's be honest, it'd be pretty weird if the answer wasn't the root 1, so let's prove that. If $\sigma(\alpha) = \alpha$, then $\mathrm{Tr}(\alpha) = \sum_{i=0}^{p-1} \sigma^i(\alpha) = p\alpha = 0$. Since we are not in characteristic p, this implies $\alpha = 0$. Thus, σ has no eigenvectors of eigenvalue 1. Thus, the characteristic polynomial is $\frac{x^p-1}{x-1}$.

9. (January 2022 Problem 2) For each of the following statements, either prove it or provide a counterexample.

(a) Every matrix of finite order in $\mathrm{GL}_n(\mathbb{C})$ is diagonalizable.

(b) Every matrix of finite order in $\mathrm{GL}_n(\mathbb{Q})$ is diagonalizable.

(c) Every matrix in $\mathrm{GL}_n(\overline{\mathbb{F}}_p)$ has finite order. (We write $\overline{\mathbb{F}}_p$ for the algebraic closure of the finite field $\mathbb{F}_p$.)

(d) Every matrix of finite order in $\mathrm{GL}_n(\overline{\mathbb{F}}_p)$ is diagonalizable.

9. Solution

(a) Suppose $A \in \mathrm{GL}_n(\mathbb{C})$ has finite order, so that $A^k = I$ for some k. Then we know the minimal polynomial of A divides $x^k - 1$. Now, over $\mathbb{C}$, $x^k - 1$ splits into k different linear factors, so the minimal polynomial also splits into different linear factors. Since the exponents on the linear factors of the minimal polynomial are the sizes of maximal Jordan blocks, this tells us that the size of a maximal Jordan block is 1, which is the same as saying A is diagonalizable.

(b) Everyone's favorite counterexample:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

(c) Let $\mu_A(x)$ be the minimal polynomial of A. Let $\mathbb{F}_q \supset \mathbb{F}_p$ be the smallest extension containing all the roots of $\mu_A(x)$. Then all the roots of $\mu_A(x)$ are nonzero elements of $\mathbb{F}_q$, and so satisfy $\alpha^{q-1} = 1$. Since $\mu_A(x)$ factors over $\mathbb{F}_q$, write

$$\mu_A(x) = (x - \alpha_1)^{e_1} \cdots (x - \alpha_k)^{e_k}.$$

If all the e_i were 1 then we would just know that $\mu_A(x) \mid x^{q-1} - 1$, and we would be done, but if one of the $e_i > 1$ then we want to find another polynomial of the form $x^{\text{something}} - 1$ which μ_A divides. We know $x^{q-1} - 1$ has distinct roots, so if $e = \max_i(e_i)$, then $\mu_A(x) \mid (x^{q-1} - 1)^e$, but this isn't in the form we want yet. To get it in the desired form, we can instead choose the smallest power of p which is greater than e. Namely, set $m = \log_p(e)$, then we have $\mu_A(x) \mid (x^{q-1} - 1)^{p^m} = x^{p^m(q-1)} - 1$. Thus, $A^{p^m(q-1)} = I$. (I actually think this exponent might be optimal, but I'm not sure. Lemme know if you figure out one way or the other.)

(d) The problem with our proof from part (a) is that we said $x^k - 1$ splits into *distinct* linear terms over $\mathbb{C}$, but that need not be true in $\overline{\mathbb{F}}_p$. For example, $x^p - 1$ definitely doesn't.

So, take the example of

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

over $\overline{\mathbb{F}_2}$.